Scientific Research Methodology

7. Confirmation bias: what is it and how can we avoid it?

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Introduction

What is a "bias"? A systematic deviation from rational assessment, evaluation, or judgement.

⋆ "bias" = partiality

⋆ "confirmation bias" = a preference for arguments or evidence that confirm one's own ideas

⋆ the tendency to seek, interpret, favour, and remember information in ways that confirm pre-existing hypotheses or beliefs

Question: Have you ever started a project hoping that the result would be X?

Psychological foundations

Why is confirmation bias a natural tendency?

- **Cognitive economy:** the brain prefers shortcuts
- **Emotional comfort:** information consistent with pre-existing beliefs and knowledge feels safer
- **Pattern detection:** people tend to see patterns or structures even in random situations
- **Example:** A researcher believes that stress and productivity are correlated: they will feel more satisfied when they find papers that confirm this view

System 1 and System 2

Two models of thinking proposed by Daniel Kahneman

- System 1: fast, automatic, intuitive, emotional
  - handles quick judgements and simple tasks
  - requires no intellectual effort
  - Examples: recognising faces, driving on an empty road
  - operates continuously, creating impressions, feelings, and inclinations without conscious control
- System 2: analytical thinking
  - deep, goal-directed thinking; requires intellectual effort; logical, conscious, slow
  - complex calculations, careful planning
  - Examples: parking in a tight space, comparing complex options, learning a new procedure or concept
  - requires attention and focus; feels mentally demanding

System 1 and System 2: implications

- Some concepts, procedures, and techniques move into System 1 after spending **enough time in System 2**: the role of school education
- We tend to use System 1 reflexively in many situations
- Scientific research requires System 2: we are facing a new situation
- **Confirmation bias**: encourages us to relax our thinking by relying on System 1

Confirmation bias in research: the literature

- selecting and reading only papers that confirm the desired hypothesis
  - **Experiment:** Search online for evidence supporting a claim, then for evidence against it. In many cases, you will find convincing resources for both searches. How do we decide what view to form?
  - a researcher must remain open about the **research outcome**: **truth matters more than sensationalism**
- ignoring studies that contradict the desired hypotheses or conclusions
- valuing "highly cited" papers when they fit our expectations

Confirmation bias in research: formulating hypotheses

- starting with the desired conclusion and constructing hypotheses by reasoning backwards
- conducting research to "obtain a particular result"
- neglecting whether hypotheses are true if they support the desired thesis

Confirmation bias in research: experimental design

- designing experiments that can only confirm the desired hypotheses
- choosing variables that make positive results more likely
- stopping an experiment when the results "look good"

Confirmation bias in research: experimental design

- recording only data that match the desired outcome
- treating anomalies differently depending on whether they align with expectations

Examples:

1. Selecting a study variable: a simple question for a complex concept; choosing an irrelevant variable that will respond positively
2. Asymmetric conditions: experimental groups are treated differently in ways beyond those stated in the study objective
3. stopping the study when the objective appears to have been reached
4. testing only cases in which success is assured
5. suggesting the desired answer in a survey
6. failing to use randomisation

Confirmation bias in research: data analysis

- "P-hacking": trying models, tests, and transformations until a significant result is obtained
- Interpretive preference: overvaluing small positive results; disregarding negative results

P-hacking—Example

- **Hypothesis:** A new supplement improves reaction time
- **In reality:** the supplement has no effect
- Data are collected and different tests run. All but one yield statistically non-significant results: $p > 0.05$
- Only the test that confirmed the desired result is retained
- **Alternatives:** change the outcome measure (mean, median, top 10% of the data, removing outliers, using subgroups)

Confirmation bias in research: peer review and publication

- reviewers favour results that confirm general expectations
- journals show the same preference

Example 1: medical research

Hypothesis: Drug X helps improve condition Y.

- reporting only tests that yield positive results
- excluding negative results on "methodological" grounds
- running different subgroup analyses until a significant result is found

Other examples

Computational experiments:
⋆ Trying many parameters until a simulation confirms the hypothesis
⋆ not reporting unsuccessful simulations

Social science surveys:
⋆ asking questions that increase the likelihood of confirming the assumed hypothesis

How can we reduce confirmation bias?

A. Preregistration: write down the hypotheses, experimental design, variables, and analysis plan **before** conducting the study
B. Critical thinking in a team: ask colleagues to challenge the study hypothesis; encourage criticism before running experiments
C. A falsification mindset: seek ways in which the hypothesis could be shown to be wrong. Example: "If my hypothesis were wrong, what would the data look like?"
D. Blinded analysis: hide group labels until outcomes and statistical models have been selected and applied
E. Structured literature review: decide on exclusion criteria before starting

Checklist

1. Have I looked for evidence that contradicts the hypothesis?
2. Have I explored alternative hypotheses?
3. Is my analysis method **fixed** or **flexible**?
4. Would someone with the opposite expectations agree with my interpretation?
5. Have I documented all experiments, or only the "good" ones?

Example A

In a study of 60 patients, those who received treatment A had slightly improved symptoms compared with the control group (but the difference was not statistically significant).

- how is this paragraph interpreted from two starting points: by someone who believes the treatment works and by someone who does not?

Conclusion

Confirmation bias: take care not to reinforce your prior beliefs automatically

⋆ use a checklist to check that you are taking an appropriate perspective

⋆ References:

- Kahneman—Thinking, Fast and Slow
- Nickerson—"Confirmation Bias: A Ubiquitous Phenomenon in Many Guises" (1998)
- Ioannidis—"Why Most Published Research Findings Are False"